AP Precalculus **Unit 4, Lesson 4.5**
Homework — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Practice

1. Consider the equation $x^2 + y^2 = 49$.

   (a) Solve for y. Write the two explicit functions.

   $f_1(x) = \sqrt{49 - x^2}$ $\quad$ $f_2(x) = -\sqrt{49 - x^2}$

   (b) State the domain of each function.

   Domain: $[-7, 7]$ **for both**

2. Complete the table for the upper half of $x^2 + y^2 = 49$.

x	$y = f_1(x)$
-7	0
0	7
5	$\sqrt{24} \approx 4.899$
7	0

3. Two points: $(5, \sqrt{24})$ and $(6, \sqrt{13})$.

   (a) $\Delta y/\Delta x \approx (3.606 - 4.899)/(6 - 5) = -1.293/1 \approx -1.29$

   (b) Co-variation: **Negative. As x increases, y decreases (opposite directions).**

4. Budget constraint: $3x + 5y = 60$.

   (a) $y = (60 - 3x)/5 = 12 - 0.6x$

   (b) **As x increases by 1 notebook, y decreases by 0.6 binders (since the coefficient of x is -0.6).**

   (c) Co-variation: **Negative**

5. Equation $9x^2 + y^2 = 9$. Upper half, x from 0 to 1.
   At $x = 0$: $y^2 = 9$, so $y = 3$. At $x = 0.5$: $y^2 = 9 - 9(0.25) = 6.75$, so $y \approx 2.598$.
   $\Delta y/\Delta x \approx (2.598 - 3)/0.5 \approx -0.804$.
   Co-variation sign: **Negative.** Reason: **As x increases from 0, y decreases from 3, so the ratio is negative.**

AP-Style Multiple Choice

6. The equation $x^2 - 4y^2 = 16$. Upper branch $(y > 0)$, x from 4 to 8.
At $x = 4$: $16 - 4y^2 = 16 \Rightarrow y = 0$. At $x = 8$: $64 - 4y^2 = 16 \Rightarrow 4y^2 = 48 \Rightarrow y = \sqrt{12} \approx 3.46$.
As x increases from 4 to 8, y increases from 0 to ≈ 3.46. Same direction $\Rightarrow$ positive co-variation.

(A) Co-variation is positive because y increases as x increases. $\leftarrow$ **correct**

(B) Co-variation is positive because y decreases as x increases.

(C) Co-variation is negative because y increases as x increases.

(D) Co-variation is negative because y decreases as x increases.

Note: On the upper branch of a hyperbola $x^2 - 4y^2 = 16$, as x moves away from the vertex at $(4, 0)$, y increases. This is opposite behavior from the circle, where the upper half has negative co-variation.

Extension optional

Extension (optional): $(x-2)^2 + (y-1)^2 = 25$ is a circle with center $(2, 1)$ and radius 5.
Explicit functions: $f_1(x) = 1 + \sqrt{25 - (x-2)^2}$ (upper), $f_2(x) = 1 - \sqrt{25 - (x-2)^2}$ (lower).
Near $x = 2$: At $x = 2$, $y = 1 + 5 = 6$. At $x = 3$, $y = 1 + \sqrt{24} \approx 5.899$. $\Delta y/\Delta x \approx (5.899 - 6)/1 \approx -0.1$. Negative co-variation.

Teacher Note

Problem 4 (budget constraint): This is a linear implicit equation. Co-variation is always negative here (constant rate -0.6). Connect to the constant rate of change for linear functions.

Problem 6 (MC): The hyperbola's upper branch behaves differently from the circle's upper half. Students who overgeneralize from the circle may choose (C) or (D). Encourage checking with two actual points.

Common error across all problems: Students may compute $\Delta x/\Delta y$ instead of $\Delta y/\Delta x$. Remind them: numerator is the variable whose change we're measuring (output in context).